

MLOps: Machine Learning Operations

Session 1

Foundations, Personas & the MLOps Lifecycle

Juan Manuel Ramírez García · jramirezg@faculty.ie.edu

Live online session · Saturday 3 October 2026 · 90 minutes

Data ingestion &
versioning

Data validation

Data pre-processing

Model training

Model tuning

Model analysis

Model validation

Model deployment

Model feedback

Today in one slide: what you will be able to do at 11:30

By the end of this session you can...



Explain why MLOps exists

Say in plain words what MLOps is and why trained models fail to deliver value.



Name who does what

Describe the personas of an MLOps team and where they hand over to each other.



Walk the lifecycle

Follow a model from data to feedback and tell "training" from "operationalising".

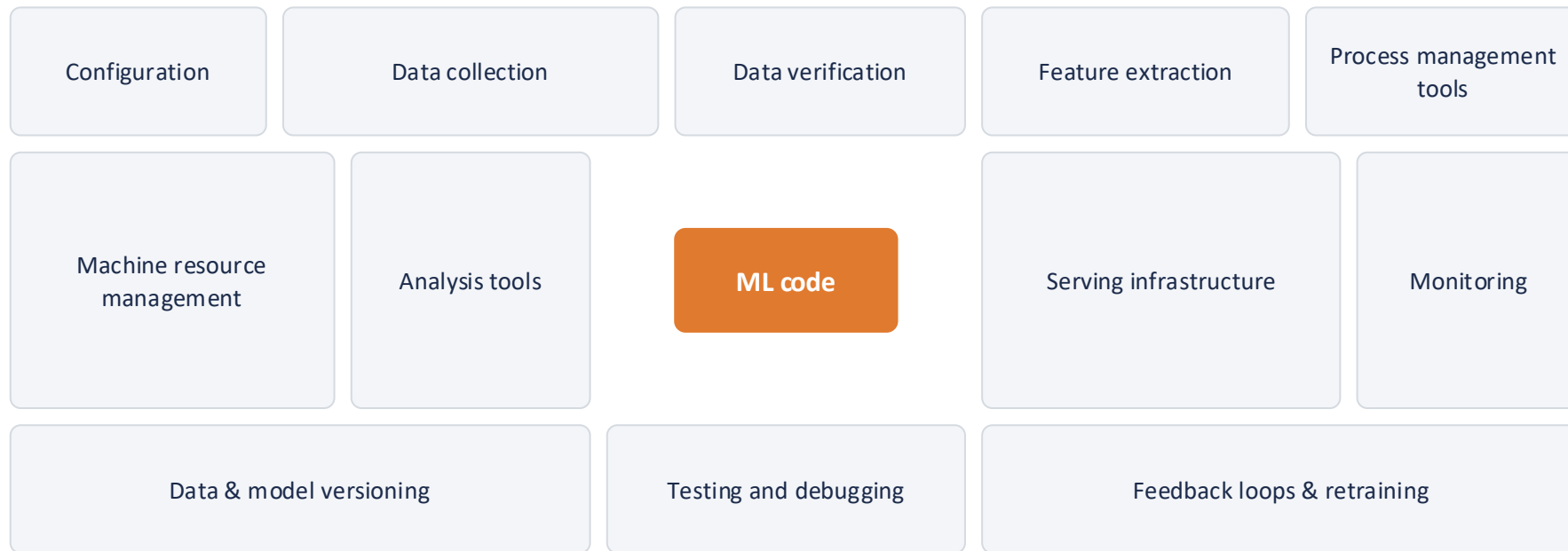


Know how the course works

Sessions, forums, videos, assessment and the AI policy.

Minute	Block
0–8	Welcome, outcomes, why this course exists
8–16	How this course works
16–35	What MLOps is and why (+ 4-min demo, 1 poll)
35–46	Who does what: personas
46–51	Break
51–58	The lifecycle and the four kinds of performance
58–76	Activity: diagnose a failed deployment
76–86	Debrief
86–90	Three things, your forum, your reading

The model is the small box in the middle



Adapted from Sculley et al., "Hidden Technical Debt in Machine Learning Systems", Google, 2015

Training a model is the first mile, not the finish line

28%

of AI use cases fully succeed and meet ROI expectations; 20% fail outright

*Gartner survey of 782 I&O leaders, Nov–Dec 2025
(published April 2026)*

95%

of enterprise GenAI pilots show no measurable impact on profit and loss

MIT NANDA, "The GenAI Divide", 2025

87%

of data-science projects never make it into production

VentureBeat, 2019 (the historical baseline)

 **The pattern has held for seven years. The problem is not the algorithms; it is everything around them.**

How this course works

Twelve sessions in five weeks: live, forum, video

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The lifecycle we follow through the course · highlighted: where this section sits

The rhythm: a live session every Saturday, a forum every week, two videos

Sat 3 Oct S1 Live Foundations, personas, lifecycle	Sat 10 Oct S3 Live Pipelines, environments, security	Sat 17 Oct S5 Live CI/CD & deployment strategies	Sat 24 Oct S7 Live Monitoring, XAI, Responsible AI	Sat 31 Oct S10 Live LLMOps & failure diagnostics
S2 Forum 1 · 3–9 Oct	S4 Forum 2 · 10–16 Oct	S6 Forum 3 · 17–23 Oct	S8 Forum 4 · 24–30 Oct S9 Video 1 · Monitoring metrics	S11 Forum 5 · 31 Oct–6 Nov S12 Video 2 · AI governance

 Live online session (90 min): all new concepts are introduced here

 Forum (Sat–Fri): a recap of the live session in three questions · Video: self-paced, with a quiz

Final examination at the end of the term.

How you are assessed: four components, all tied to what we practise



35% · Final exam

- Individual, end of term
- Lifecycle, pipelines/CI-CD, monitoring and drift, governance
- One handwritten A4 sheet allowed
- Minimum 3.5/10 to pass the course



35% · Group project

- Teams of 3–5 as an MLOps consulting team
- Roadmap, personas, CI/CD schematic, monitoring policy, governance
- Defended in front of the class



15% · Quizzes

- Short quizzes across the modules
- Includes the two knowledge checks inside Video 1 and Video 2



15% · Participation

- Live sessions and forums
- Judged on quality and insight, not volume
- Connect concepts to real examples



Everything you are assessed on is something you practise first: in the live activity, then in the forum.

How to do well in the forums, and how to use AI tools honestly



Quality, not volume

- A good post connects a concept from the session to a real example and a constraint
- Reply to others: challenge, extend, or ask for evidence
- Avoid being the Repeater (restating points), the Rambler (long tangents) or the Distracted (asking what was already answered)
- Post early in the week; the best debates need time



Generative AI: encouraged, and declared

- Use it to draft, prototype pipelines, structure arguments
- Not in the final exam or in quizzes
- Add to every submission: "I acknowledge the use of [AI system] to [task]. Prompts used: [sequence]."
- Or: "No content generated by AI has been used in this assignment."
- Unattributed use is academic misconduct

The course follows the life of a model: four modules, one lifecycle

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Module 1 · Sessions 1–2 **Foundations & lifecycle planning**

Why MLOps, the personas, the end-to-end lifecycle. Today.

Module 2 · Sessions 3–6 **Pipelines, environments & deployment**

From notebook to pipeline; Dev/QA/Prod; security; CI/CD; rollout strategies.

Module 3 · Sessions 7–9 **Monitoring, reliability & Responsible AI**

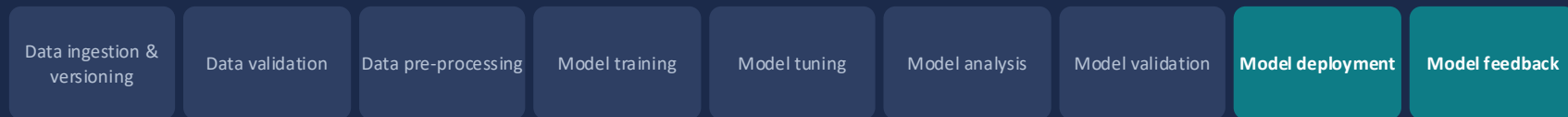
Drift, decay, retraining loops, explainability, bias. Video 1: metrics in depth.

Module 4 · Sessions 10–12 **GenAI operations & governance**

LLMOps, RAG, failure diagnostics. Video 2: EU AI Act, NIST, audit readiness.

What is MLOps, and why does it exist?

From DevOps to MLOps: models are not static



The lifecycle we follow through the course · highlighted: where this section sits

Software solved this first: DevOps, and its two words, CI and CD



DevOps: a mindset and practices

Developers and operations collaborate; the time from a change to that change running reliably in production shrinks.



Continuous Integration (CI)

Every change is merged into the shared version and checked automatically. Needs version control.



Continuous Delivery (CD)

A build that passed the checks goes to a staging environment, is tested, then to production, automatically.



An assembly line: every part is checked as it arrives (CI) and a car only leaves when it has passed every station (CD). Session 5 adds a third word for ML: Continuous Training.

The extra problem with ML: the world keeps changing under the model



Credit model, banking

- Customer behaviour changes
- Customer base changes
- Bias appears or grows



Recommendations, TV platform

- Viewer tastes change
- The catalogue changes every week
- New competitors change what "good" means



Sales forecast, retail

- Consumer behaviour changes
- Economic situation changes
- Promotions and seasons move



A program that calculates VAT works the same in ten years. A model learned from the past, and the past stops being true. ML models need retraining: not if, but when.

Why managing models at scale is hard



Many dependencies

Data sources, libraries, infrastructure, other models.



Data keeps changing

What was true when the model was trained is not true now.



Business needs shift

The goal the model was built for moves.



Not everyone speaks the same language

Business, data, modelling and IT each have their own vocabulary.



Data scientists are not software engineers

Trained to build models, not to run services.



Complexity grows faster than the model count

Many models in production; staff turnover; nobody remembers why.



Chat poll (2 min): "MLOps is an unnecessary cost for a company with only three models in production." AGREE or DISAGREE, and one line why.

So, what is MLOps?

MLOps is the engineering discipline that bridges the gap between data-science experimentation and reliable software production. It standardises how models are deployed, scaled, monitored and governed so that they stay accurate, resilient and audit-ready over time.



Deployed and scaled

The model runs where the business needs it, at the volume the business needs.



Monitored

Someone (or something) notices when it stops being right.



Governed

You can prove what it does, why, and who approved it.



A model in production is a running process, not a file.

Training a model is not the same as operationalising it



Training a model

- Ends with a file and a score on historical data
- One person, one laptop, one moment in time
- Success = the metric looks good
- Owner: the data scientist



Operationalising a model

- The model answers real requests every day, with today's data
- Runs on infrastructure, integrated in an application
- Success = the business metric moves and stays moved
- Owner: a team, with someone accountable



Consequence of confusing them: a demand-forecast model is declared "done" in January because accuracy was 92% on last year's data. Nobody owns it after that. By June it is silently wrong and the cost is measured in write-offs.

One habit underneath all of this: every change is recorded and can be undone



A repository is a folder with a memory

Every change is a commit: who, when, and why.



You can compare any two moments

A "diff" shows exactly what changed between two versions.



You can name a moment and go back to it

A "tag" says: this is the version that went to production on 3 March.



Live demo · 4 minutes

A real repository of a small ML project: the commit history, one diff, the tags. No commands, just what "versioning" looks like.

What you just saw



The history

A list of commits: who changed what, when, and the message that says why.



The diff

Exactly which lines changed between two moments. This is what a reviewer checks before a change is accepted.



The tag

A named version. "Which model was live in March?" has an answer.



In Session 3 we extend the same idea to data and models: versioned datasets, an experiment log and a model registry.

Who does what: the MLOps personas

Roles, duties and the hand-overs between them

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The lifecycle we follow through the course · highlighted: where this section sits

Six personas, six different questions about the same model



Business / Product owner

"Is it worth it, and is it useful?"

Defines the goal, the product metric and the guardrails; decides what "good enough" means.



Data scientist

"Is it accurate?"

Explores data, builds and evaluates the model; hands it over with documentation.



Data engineer

"Is the data available, correct, on time?"

Builds the data pipelines and curated datasets the model depends on.



ML engineer (MLOps)

"Does it run reliably in production?"

Packages, deploys, automates retraining and monitoring; the bridge between DS and IT.



IT / DevOps

"Does the infrastructure hold and stay secure?"

Runs the platform: capacity, scaling, access, uptime.



Risk & Compliance

"Can we prove it is fair, explainable and allowed?"

Assesses model risk, audits, documentation, regulatory obligations.

Based on Treveil et al., Introducing MLOps (O'Reilly, 2020), with Product Owner and Risk & Compliance made explicit

Who talks to whom: the nine hand-overs of a model



Adapted from Treveil et al., Introducing MLOps (O'Reilly, 2020)

A tool for the forum: the RACI map

R • Responsible

Does the work.

A • Accountable

Owns the outcome and signs off. Only one A per row.

C • Consulted

Gives input before the decision.

I • Informed

Is told after the decision.

Lifecycle stage	Product owner	Data scientist	Data engineer	ML engineer	IT / DevOps	Risk & Compl.
Data ingestion & validation	I	C	R / A	C	I	C
Model development	C	R / A	C	C		I
Model deployment	A	C	I	R	C	I
Monitoring & retraining	I	C	C	R / A	C	C

An example, not the answer: the forum asks you to argue your own allocation.

Break · 5 minutes

Back at minute 51. While you wait: which of the six personas is closest to your current or future job?

The end-to-end lifecycle, and what "performance" means

Nine steps, four kinds of performance, one plan

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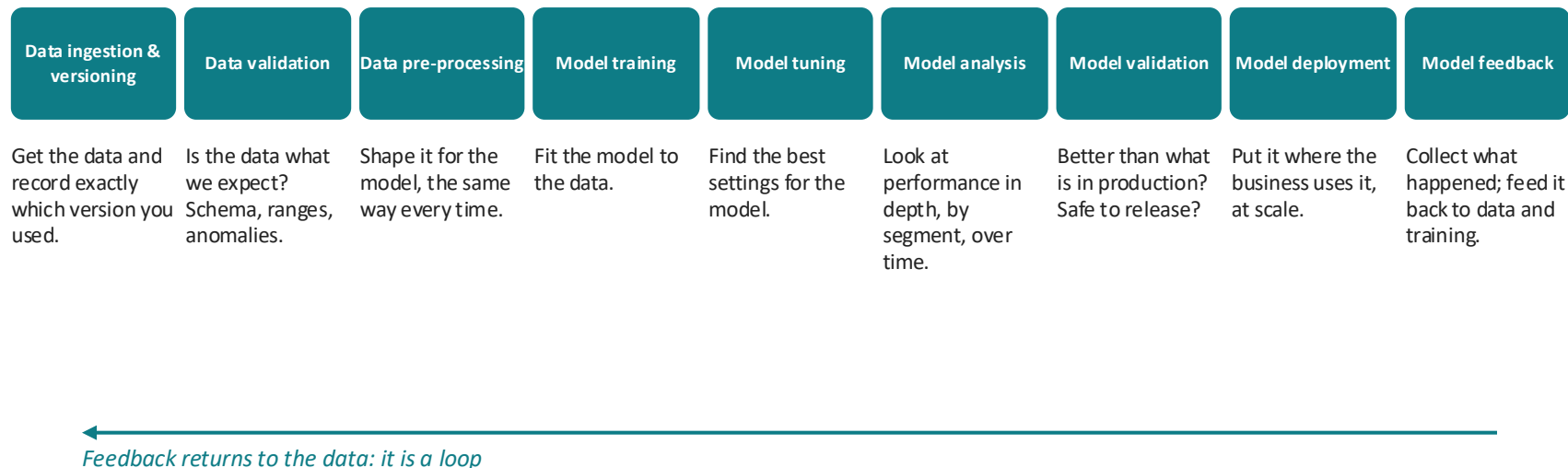
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The lifecycle we follow through the course · highlighted: where this section sits

The lifecycle in plain words: nine steps from data to feedback



The first three steps are about data, not models. In practice that is where most of the time, and most of the failures, live. Session 3.

Lifecycle from Hapke & Nelson, Building Machine Learning Pipelines (O'Reilly, 2020)

"Performance" means four different things, and you must define all four



Business performance

- Goal → product metric (click-through rate, users attracted)
- Guardrail metric that must not fall (session length)
- The only metrics that matter in the end



Model performance

- Accuracy, AUC, F1, RMSE... chosen for the problem
- Sufficient for the goal, not the maximum
- A product that needs a perfect model will fail



Data performance

- Will tomorrow's data look like the training data?
- How often must we retrain, and what does it cost each time?
- Some problems need daily freshness, others yearly



System performance

- How fast is fast enough? At what cost per prediction?
- Hardware limits, development time, maintainability
- Never confuse performance in development with production



Most failures are decided here, before training: a product metric and a model metric pointing in different directions, or an accurate model nobody acts on.

Activity · 15 minutes in breakout rooms: diagnose a failed deployment



Your task (15 min)

- Read the one-page case: NorthRetail demand forecast
- Four things went wrong. For each: which lifecycle stage did it come from? which persona should have caught it? which single control would have prevented it?
- Bonus: what should IT's dashboard have shown?



How to work

- Rooms of 4–5: your future project team
- Fill the diagnostic template from the campus (or a whiteboard)
- Nominate a spokesperson: 2 minutes to present
- Use the vocabulary from today: the nine lifecycle steps, the six personas



Output per room: one slide or whiteboard. Three rooms present in the debrief; everyone posts theirs on the forum.

The case: NorthRetail demand forecast (facts only)

- January: a data scientist delivers a demand-forecast model for 400 stores. Accuracy on last year's data: 92%. The project is declared complete.
- February: a new supplier system changes the product codes. The model keeps running; the affected features arrive empty, and the model treats them as zero.
- March: the commercial team moves to weekly promotions instead of monthly. Nobody tells the data team.
- April: the data scientist leaves the company. The model has no documented owner.
- January–June: IT's dashboard shows the forecasting service "green" every day: it is up, and it answers in 40 ms.
- June: out-of-stock incidents are up 18%; overstock write-offs reach €2.1M. The model is switched off and replaced by last year's manual spreadsheets.

Diagnostic template

- What went wrong (4 items)
- Lifecycle stage
- Persona who should have caught it
- One control that would have prevented it
- Bonus: what should the IT dashboard have shown?

Three things to remember

1

A model in production is a running process, not a file.

Training ends with a file and a score. Operationalising means it answers real requests every day, and someone is accountable.

2

The world changes under the model.

Unlike ordinary software, models decay as data and business change. Retraining is a question of when, not if.

3

Six personas, nine hand-overs.

Most failures happen between roles, not inside them. Decide who is Responsible and who is Accountable for each stage before you start.

Your forum this week: Session 2 · 3–9 October

Q1

Choose a real or hypothetical company and explain, in a short post, three reasons its ML models might never reach production. Which persona is best positioned to address each?

Q2

Draft a one-page responsibility map (RACI-style) for a fictional recommendation-model project, assigning the Data Scientist, ML Engineer, IT/DevOps, Product Owner and Risk/Compliance roles across the lifecycle stages.

Q3

In your own words, define the difference between "training a model" and "operationalising a model", and give one concrete business consequence of confusing the two.



Opens today, closes Friday 9 October. Post early; reply to at least two classmates; my synthesis goes up on Friday.

Before Session 3 (Saturday 10 October): read, and look at one thing



Read

- Treveil et al., Introducing MLOps (O'Reilly, 2020), chapters 1–3: Why now, People of MLOps, Key features
- Optional: Huyen, Designing Machine Learning Systems (2022), chapter 1



Look at one thing

- On campus: the exploratory notebook (PDF, nothing to run)
- Write down three things that look messy or fragile
- Session 3 starts from your list

Next: Session 3 · ML Pipelines, Environments & Security

From notebook to pipeline · why Dev, QA and Prod are separate · how a model can be attacked, and how to protect it